

Suppression of the quadrupole-misalignment false EDM signal by orbit-based methods in a proton EDM storage ring, and its limits

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The proposed storage-ring experiment for the proton electric dipole moment (EDM) must detect a spin rotation of about one nanoradian per second. At that level, quadrupole magnets displaced from their ideal positions by only a few microns produce a spin signal that imitates a real EDM. This “false EDM” comes from a geometric (Berry) phase and grows as the *square* of the misalignment, so at a realistic alignment of $10\ \mu\text{m}$ it is about a thousand times larger than the target. We ask a simple question with a simulation of the ring: how much of this false signal can be removed by measuring and correcting the beam orbit, the everyday toolbox of accelerator alignment? Standard orbit correction, together with running clockwise and counter-clockwise beams at once, removes a factor of about 26, but no further. The reason is that the leftover misalignment lives in a special pattern — neighboring quadrupoles displaced *together* — whose effect on the orbit is roughly a hundred times weaker than for the opposite pattern. We show that a whole family of orbit techniques (matrix inversion, per-magnet gradient modulation read as an amplitude, neural networks, drift monitoring, harmonic fits, and the counter-rotating beam separation) all stall at the same floor, and that the floor is set not by detector noise — which can be averaged away — but by two coherent effects that cannot: the optics “breathing” when a magnet is modulated, and the mismatch between the real optics and the model used to analyze it. Better beam-position monitors, including SQUID-based ones, do not move this floor. We then identify one technique that escapes it: null-finding beam-based alignment (BBA), which locates each magnet center by the beam position where the orbit stops responding. A zero-crossing does not care about the response size, so it is not blocked by these walls. In idealized optics it recovers the hidden pattern and cuts the false EDM from $356\times$ to $1.6\times$ target. Under realistic optics errors it works in the vertical plane after the orbit is first flattened and the procedure iterated, but stalls in the horizontal plane because our steering model treats the electric deflectors as field-free drifts, which is accurate vertically but not horizontally. We present this as the one promising — but not yet finished — route, and conclude that reaching the target by the other techniques would require the symmetric misalignment to be held below about $1\ \mu\text{m}$ by construction, a tolerance that no orbit measurement can verify.

I. INTRODUCTION

A. The experiment: searching for a proton EDM

An electric dipole moment (EDM) is a tiny separation of a particle’s positive and negative charge along its spin. If it exists, it breaks time-reversal symmetry, and finding one in the proton above the Standard-Model level ($\lesssim 10^{-31}\ \text{e cm}$) would be direct evidence of new physics. The proposed storage-ring experiment [1] aims for $d = 10^{-29}\ \text{e cm}$.

The method is called *frozen spin*. Protons circulate at a special “magic” momentum ($p \approx 0.7\ \text{GeV}/c$) in a ring that bends the beam with electric fields and focuses it with magnetic quadrupoles. At this momentum the spin stays pointed along the direction of motion; it does not drift ahead or behind within the ring plane. If the proton has an EDM, the ring’s radial electric field slowly tips the spin *out* of that plane. The measured quantity is the rate of that tip,

$$f \equiv \frac{dS_y}{dt}, \quad d = 10^{-29}\ \text{e cm} \leftrightarrow f \approx 1\ \text{nrad/s}, \quad (1)$$

one radian in about thirty years. Every other effect that can tip the spin out of plane at this level must be removed

or safely bounded. Effects that imitate the EDM are called *false EDM* signals; this paper is about the most stubborn one.

B. The false EDM: a geometric phase from misalignments

The ring has 48 quadrupole magnets, and in practice each sits a few microns away from its ideal place. A quadrupole has zero field at its center, so a beam passing off-center feels a field proportional to how far off it is. Which way the magnet is displaced decides what it does to the spin. A *vertical* offset dy gives the beam a radial field that rotates the spin about the horizontal (x) axis. A *horizontal* offset dx gives it a vertical field that rotates the spin about the vertical (y) axis.

Either rotation on its own is nearly harmless, because it averages out around the ring. Trouble comes when both are present at once. Rotations about different axes *do not commute*: rotating first about x and then about y is not the same as doing them in the other order, and the small leftover difference builds up turn after turn into a net rotation about the third axis — exactly the out-of-plane direction where the EDM is sought. This is the

familiar geometric (Berry) phase. Its size is

$$f_{\text{false}} \propto dx \cdot dy \implies f_{\text{false}} \propto \sigma^2, \quad (2)$$

so it depends on the *product* of the two offsets, and therefore grows as the *square* of the rms misalignment σ . We verify the square law directly in Sec. II B (measured exponent $p = 2.00 \pm 0.01$, matching Fig. 9(a) of Ref. [1]). The number is unforgiving: at $\sigma = 10 \mu\text{m}$ the false EDM is about $1000\times$ the target, and it lives in the *same* measured quantity, dS_y/dt , as the real signal.

The obvious hope is that the beam itself can be used as the ruler. The ring’s 48 beam-position monitors (BPMs) continuously read where the beam sits; misaligned quadrupoles bend that path; so measuring the misalignments from the orbit and correcting them — the standard *beam-based alignment* (BBA) practice — ought to clean up the false EDM as well. This paper measures how far that hope holds, and where and why it stops.

C. What is already known

Most pieces of this problem are, individually, solved. Our contribution is to put them together around the false-EDM question.

a. Orbit correction and observability. Correcting the orbit by singular-value decomposition (SVD) of the response matrix — including the idea of poorly-visible modes and how to reject them — goes back to Chung, Decker, and Evans [3]. The gain law we use throughout, $G_k = C/|Q^2 - k^2|$, is the harmonic limit of that picture. The bias of estimators when the orbit-response inversion is nearly degenerate was worked out by Wegscheider and Vilsmeier [4]; their degenerate parameter is the quadrupole strength, ours the transverse offset.

b. Beam-based alignment. Steering the beam through quadrupole centers is routine: vary a quadrupole’s strength and find the beam position where the orbit stops responding, speed it up with ac excitation [5], and do many magnets at once [6]. None of the measurement techniques we test is new. *What is new is measuring how well each one does on the specific misalignment pattern that drives the false EDM.*

c. False-EDM budgets. The forward calculation, misalignment $\rightarrow$ false EDM, was done for an all-electric ring in Ref. [2]. The full systematic budget of the magnetic-quadrupole symmetric-hybrid design is given by Omarov *et al.* [1], who show the geometric-phase false EDM can be brought below target by combining clockwise (CW) and counter-clockwise (CCW) beams, quadrupole polarity switching, and shrinking the separation between the two counter-rotating (CR) beams with dipole correctors.

d. Neighboring work. Orbit correction in symmetric lattices [7]; updating the response matrix on line [8]; the orbit imprint of ground motion [9]; polarimeter performance and statistics [10].

D. The open gap

The control chain of Ref. [1] relies on *measuring* the CR beam separation and reducing it. Three things are left open there. (i) The measurement chain itself — 48 BPMs (or SQUID pickups) plus the reconstruction — is not simulated. (ii) A possible blindness to the part of the misalignment that the orbit cannot see is not addressed. (iii) Whether polarity switching is a separate handle from CW/CCW reversal is not quantified. These matter because, as we show, the part of the misalignment that drives the residual false EDM is exactly the part the orbit sees least.

E. This paper

With realistic BPM systematics (offsets, gain errors, noise) and optics-model errors (β -beating, quadrupole tilt), we find the false-EDM floor reachable by orbit measurement and correction, as a function of the alignment tolerance. At $\sigma = 10 \mu\text{m}$ the floor is $\sim 6 \times 10^{-28} \text{ ecm}$, and it scales as σ^2 . We show the floor is *independent of BPM technology*: it is set not by random noise but by two coherent effects that do not average down. And we identify one technique — null-finding BBA — that is not subject to these walls, map how far it currently reaches, and state honestly where it stalls.

Section II describes the simulation, the false-EDM estimator, the symmetric/antisymmetric split, and the error models. Section III first establishes the suppression that orbit control does deliver, then walks through each attempt to recover the hidden component and where it stops, and finally presents null-finding BBA. Section IV draws the conclusions for the experiment.

II. METHOD

A. Two simulation tools

Every number here comes from one of two tools that share only the configuration file, so agreement between them cannot come from a shared modeling mistake.

a. Full particle-and-spin tracker (C++). The ring — electric bending sections plus 24 FODO cells with 48 magnetic quadrupoles ($R_0 = 95.49 \text{ m}$, $g \approx 0.2 \text{ T/m}$, vertical tune $Q_y \approx 2.3$) — is pushed through the real fields element by element with a fourth-order Gauss–Legendre *symplectic* integrator, and the spin rides along in the same steps via the Thomas-BMT equation. “Symplectic” means the integrator does not let energy or amplitude errors creep up over long tracking, which is what makes a nrad/s slope measurable at all. The real EDM enters through a switch ($\eta = 1.88 \times 10^{-15} \rightarrow dS_y/dt = 9.81 \times 10^{-10} \text{ rad/s}$, odd under CW/CCW, matching Eq. (C1) of Ref. [1]). All false-EDM runs have the EDM

switch *off*, so any vertical spin rotation we see is, by construction, a systematic.

b. Analytic Twiss / closed-orbit solver. From the 2×2 transfer matrices of the quadrupoles and drifts we build the lattice functions $\beta(s)$, $\phi(s)$, the tune Q , and the orbit response matrix

$$R_{ij} = \frac{\sqrt{\beta_i \beta_j}}{2 \sin \pi Q} (KL)_j \cos(|\phi_i - \phi_j| - \pi Q) \quad (3)$$

in closed form. This runs in seconds and is used for the wide scans (noise Monte Carlo, β -beating sweeps). It agrees with the C++ tracker to within 1%. One caveat we return to in Sec. III D: this solver treats the electric deflectors as field-free drifts. That is an excellent approximation for the vertical plane, where the deflectors barely focus, but not for the horizontal plane, which is the plane the ring bends in.

c. A methodological caution. For an inverse problem the response matrix must be built from *real beam dynamics*, not from an idealized formula that can hide the least-visible mode. Here the analytic R has condition number 193, while the beam-dynamics R carries an extra near-invisible mode at the 10^{-7} level.

B. Measuring the false EDM correctly

The goal looks easy — the slope of $S_y(t)$ — but there is a trap. The beam’s betatron wiggle around the closed orbit feeds oscillations into the spin that are far larger than the slow slope we want, and a naive straight-line fit reads them as signal. Two measures are used together:

1. *Sit on the closed orbit.* The tracked particle is placed, by Newton iteration in (x, x', y, y') , exactly on the closed orbit, where there is no betatron wiggle to begin with.
2. *Fit a model, not a line.* From $S_y(t) = a + bt + \sum_k [c_k \cos \omega_k t + d_k \sin \omega_k t]$ only the slow slope b is kept; the oscillating part is absorbed by the fit. (A plain polynomial fit is *never* used.)

Three independent checks validate the estimator:

- **Square-law test:** scanning $\sigma = 2.5/5/10 \mu\text{m}$ gives $|f| \propto \sigma^p$ with $p = 2.00 \pm 0.01$ [Fig. 1] — a pure geometric phase with no linear leakage, matching Fig. 9(a) of Ref. [1].
- **Sign test:** flipping the signs of the misalignments gives $f(+dx, -dy) = -f(+dx, +dy)$ and $f(-dx, -dy) = +f(+dx, +dy)$ to the 10^{-3} level, confirming f is purely the product $dx \cdot dy$.
- **Four-particle check:** the averaged $S_y(t)$ of four symmetric starts placed *around the closed orbit* matches the single on-orbit particle to within 1%.

a. A trap worth reporting. If those four symmetric starts are placed around the *ideal axis* instead of the closed orbit, the method fails. The $\pm\delta$ wiggles cancel in pairs, but the wiggle caused by the closed-orbit offset itself is *shared* by all four particles and survives. The measurement then collapses to the same betatron artifact as a single off-orbit particle (-3.0×10^{-4} rad/s, $\sim 300\times$ the signal). Symmetric sampling does not replace knowing the closed orbit; it only works around it.

C. Two faces of a misalignment: symmetric and antisymmetric

One split underlies everything that follows. Write the 48 offsets as a vector v , with the focusing quadrupole (QF) at index $2k$ and the defocusing one (QD) at $2k+1$ in cell k . In each cell, separate how much the two magnets move *together* from how much they move *oppositely*:

$$\begin{aligned} (P_{\text{sym}} v)[2k] &= (P_{\text{sym}} v)[2k+1] = \frac{1}{2}(v[2k] + v[2k+1]), \\ (P_{\text{anti}} v)[2k] &= -(P_{\text{anti}} v)[2k+1] = \frac{1}{2}(v[2k] - v[2k+1]), \end{aligned} \quad (4)$$

with $P_{\text{sym}} + P_{\text{anti}} = I$.

Why this split matters: QF and QD have *opposite-sign* gradients. Two neighbors displaced in *opposite* directions (the antisymmetric part) kick the beam the *same* way; the kicks add up into a smooth, slowly-varying pattern near the betatron resonance, and the orbit distortion is *large*. Two neighbors displaced in the *same* direction (the symmetric part) kick *oppositely*; the kicks cancel within the cell, leaving only a fast cell-to-cell alternation at harmonic $k \approx 24$, which the orbit gain

$$G_k = \frac{C}{|Q^2 - k^2|}, \quad C \approx 24.8, \quad Q^2 \approx 5 \quad (5)$$

crushes because $Q \approx 2.3 \ll 24$ [Fig. 2, right].

This is not an assumption; it is a property of the response matrix [Fig. 2, left]. Sorting the modes of $R = U\Sigma V^T$ by how symmetric they are shows the *visible* (large-singular-value) modes to be almost purely antisymmetric and the *invisible* (small-singular-value) modes almost purely symmetric, with a visibility gap of $\text{cond}(R) = 193$. Concretely, the same $10\text{-}\mu\text{m}$ pattern leaves a $\sim 115 \mu\text{m}$ orbit trace if antisymmetric but only $\sim 1.7 \mu\text{m}$ if symmetric.

The whole tension of Sec. III follows: *the part of the misalignment that drives the leftover false EDM is the symmetric part — the part the orbit sees least.*

D. How we score a reconstruction: the correlation trap

We use two scores, and the difference between them decides several results, so we state both.

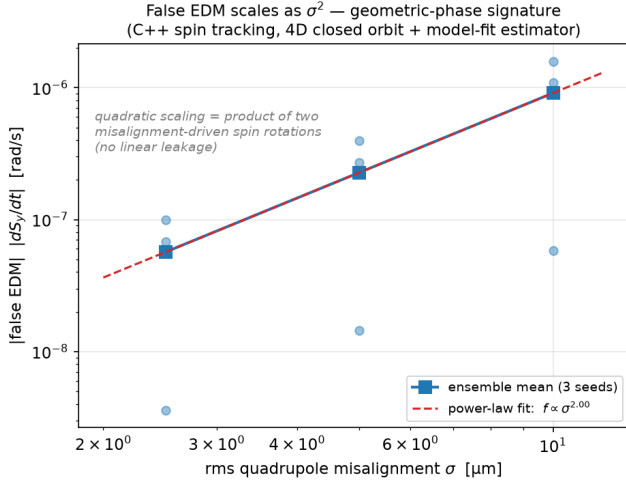


FIG. 1. Validation of the false-EDM estimator: the measured $|dS_y/dt|$ scales as $\sigma^{2.00}$ across $\sigma = 2.5\text{--}10\ \mu\text{m}$ (C++ tracking, three seeds per point) — the fingerprint of a pure geometric phase with no linear leakage. The scatter between seeds reflects that f is a product $dx \cdot dy$: individual patterns can happen to be small.

a. Pearson correlation — the misleading one. Between a reconstruction $\hat{v}$ and the truth v over 48 components, corr is the usual normalized dot product. It misleads here for two reasons. First, it subtracts the mean, which removes exactly the worst (uniform) member of the symmetric family from the error. Second, it is dominated by the many well-reconstructed antisymmetric components. As a result, $\text{corr} = 0.99$ can coexist with a completely wrong symmetric component (Fig. 4 shows a case).

b. Component error — the honest one. The rms, without subtracting the mean, of the symmetric and antisymmetric parts of the error $e = \hat{v} - v$, compared against the corresponding parts of the signal itself (both $\approx 41\ \mu\text{m}$ for $\pm 100\ \mu\text{m}$ uniform misalignments). A method truly “sees” the symmetric component only if its symmetric error is below the symmetric signal.

E. Systematic-error models

Four systematics enter at realistic sizes:

- **BPM offsets** (static, $\sim 100\ \mu\text{m}$ each; $30\ \mu\text{m}$ variant): each monitor’s electronic zero differs from the true axis by an unknown but steady constant. The main killer of single-orbit inversion; it cancels in any difference measurement (like K -modulation or drift monitoring).
- **BPM noise** (random, $\sim 1\ \mu\text{m}$ per shot): averages down as $\sqrt{N}$, so with synchronous detection it stops being a practical limit. A central message of this paper is that the real limit is *not* this.

- **BPM gain errors** (multiplicative, 1–10%): percent-level scale errors of the readings; partly softened by averaging over BPMs.
- **β -beating / model error** (per-quadrupole 0.5–5%, applied in the tracker as small gradient deviations): the gap between the real optics and the model used to analyze the data (a LOCO-calibrated machine typically leaves $\sim 1\%$). This is *coherent* — it does not average down — and it turns out to be the decisive wall.
- **Quadrupole tilt** (roll angle ψ): two channels, cross-plane leakage into the measurement and a direct geometric-phase term in the false EDM; the latter dominates and is reported as its own tolerance.

a. Seed policy. Because f is a product, it can be accidentally small or large for one misalignment draw, so every claim is averaged over many realizations (typically 3–40) and no conclusion is drawn from a single seed.

III. SIMULATION RESULTS

A. What orbit control does deliver

Start with what works. For $\sigma = 10\ \mu\text{m}$ misalignments in both planes the raw false EDM is $|f| \approx 10^{-6}\ \text{rad/s}$, about 1000 $\times$ target [Eq. (1)]. Two standard steps cut it down (Table I, Fig. 3):

1. *Run both beams at once.* The real EDM flips sign when the beam direction reverses [Eq. (C1) of [1]], so the CW/CCW difference cancels the part of the false EDM that does *not* flip. Measured gain: 3.4 $\times$ (ensemble median), leaving 474 $\times$ target.
2. *Correct the orbit.* Flattening the orbit with correctors removes the visible (antisymmetric) part of the misalignment imprint. Measured extra gain: 7.7 $\times$, leaving **62 $\times$ target** ($6.05 \times 10^{-8}\ \text{rad/s} \approx 6 \times 10^{-28}\ \text{ecm}$).

What is left after both steps sits almost entirely in the symmetric, orbit-blind pattern of Sec. II C, and by Eq. (2) it scales as σ^2 . The central deliverable of the paper is therefore the curve of Fig. 3. To reach the target along it, the *symmetric* misalignment must be at or below $\sigma_{\text{sym}} \approx 1.3\ \mu\text{m}$. Two facts make that hard. First, as the rest of this section shows, no orbit measurement can verify it. Second, a tape measure cannot verify it either, because what matters is the *magnetic* center: stray fields at the $\sim 20\ \mu\text{T}$ level move the magnetic center tens of microns away from the mechanical one, and only the beam sees the magnetic center.

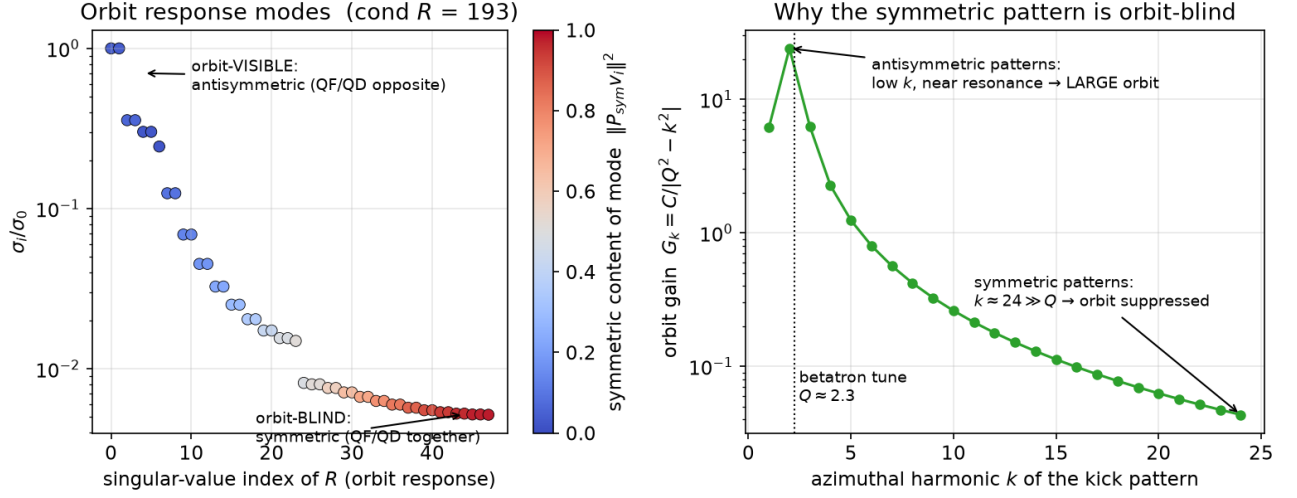
Mode structure of the orbit response (analytic lattice, $Q_y = 2.30$)

FIG. 2. Mode structure of the orbit response (analytic lattice, $Q_y = 2.30$). Left: singular values of R , colored by how symmetric the corresponding misalignment mode is; the visible modes are antisymmetric, the invisible ones symmetric; $\text{cond}(R) = 193$. Right: the orbit gain law $G_k = C/|Q^2 - k^2|$; antisymmetric patterns give low- k kicks near the resonance, symmetric patterns give $k \approx 24$ kicks far above it.

TABLE I. The suppression chain at $\sigma = 10 \mu\text{m}$ (C++ tracking). The equivalent EDM uses $1 \text{ nrad/s} \leftrightarrow 10^{-29} \text{ ecm}$.

Stage	$ f / \text{target}$	equiv. EDM [ecm]
raw (no correction)	$\sim 1000\times$	$\sim 10^{-26}$
+ CW/CCW difference	$474\times$	$\sim 5 \times 10^{-27}$
+ orbit correction	$62\times$	$\sim 6 \times 10^{-28}$
+ any method of Sec. III B	$62\times$	floor

B. Trying to measure the hidden component from the orbit

Everything in this subsection shares one idea: reconstruct the misalignments (or their symmetric part) from orbit readings, then correct. Five independent attempts hit the same wall.

1. Common-frequency K -modulation with matrix inversion

Modulate all quadrupole gradients together between two settings (g and $1.02g$) and subtract the two orbits. The static BPM offsets cancel exactly, leaving $\Delta y = \Delta R \Delta q$ with $\Delta R = R(1.02g) - R(g)$. With no errors, inverting this 48×48 matrix recovers the offsets perfectly (corr = 1.000000), and optics breathing (below) is not a problem here because it is already contained in ΔR . The problem is conditioning: $\text{cond}(\Delta R) = 3.7 \times 10^4$ with $\sigma_{\min} \approx 10^{-4}$, and the weak directions are the symmetric ones.

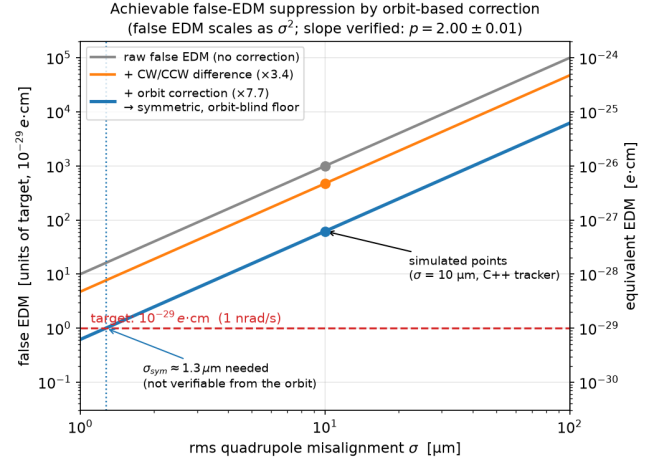


FIG. 3. Main result: the false EDM reachable by orbit correction versus the rms quadrupole misalignment. The points at $\sigma = 10 \mu\text{m}$ are C++ tracking; all curves scale as σ^2 ($p = 2.00$). The corrected floor crosses the target only at $\sigma_{\text{sym}} \approx 1.3 \mu\text{m}$ — a level no orbit measurement can verify.

Random BPM noise *can* be beaten: modulating at $\sim 1 \text{ kHz}$ with lock-in (synchronous) detection averages the noise down as $\sqrt{N}$, reaching $\sim 14 \text{ nm}$ in about 17 minutes. The limits that remain are the ones that do not average [Fig. 4]. Even at a 10 nm noise floor with a *perfect* optics model, the symmetric error is already a third of the symmetric signal. With a realistic model the inversion collapses: $\varepsilon = 0.5\%$ of β -beating drives the symmetric error to $\sim 7\times$ the signal ($283 \mu\text{m}$).

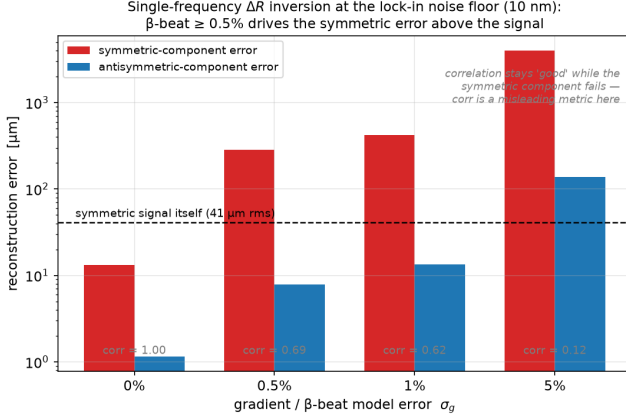


FIG. 4. Single-frequency ΔR inversion at the lock-in noise floor (10 nm, 40 seeds). Symmetric and antisymmetric reconstruction errors versus the optics-model error σ_g ; the dashed line is the symmetric signal itself (41 μm rms). Any realistic model error ($\geq 0.5\%$) pushes the symmetric error far above the signal while the correlation (gray labels) stays deceptively high — the correlation trap of Sec. IID.

vs. 41 μm), while the correlation still reads a reassuring 0.69. Because β -beating is coherent, no BPM technology — including SQUID BPMs, whose low noise would only shorten the integration time — changes this floor.

a. A lower bound behind this. This is not an accident of the scheme. Cancelling the static offsets by any two-setting difference means the useful information is no longer R but $\Delta R \sim \varepsilon_g \partial R / \partial g$, whose inverse grows as $\|\Delta R^{-1}\| \sim \|R^{-1}\| / \varepsilon_g$. So within the natural class of linear, unbiased estimators that cancel offsets by a two-setting difference, offset immunity is bought at the price of a $1/\varepsilon_g$ noise blow-up. We state this as a structural lower bound for that class; estimators outside it (e.g., Bayesian modeling of the offsets) are not covered.

2. Per-quadrupole modulation read as an amplitude: optics breathing

A seemingly stronger idea drops matrix inversion. Modulate each quadrupole at its *own* frequency ($\sim 2\%$ depth); the strength A_i of frequency f_i in a BPM signal should be proportional to that magnet’s beam offset o_i , giving a direct per-magnet readout — the ac-BBA idea of Ref. [5] moved to this ring. Because the modulation (1–10 kHz) is far below the betatron frequency (~ 515 kHz), the beam follows the slow change and each amplitude equals a static derivative $\partial y_{\text{BPM}} / \partial g_i$.

It fails for a reason worth understanding [Fig. 5]. Wiggling quadrupole i ’s strength does two things: it modulates that magnet’s own kick (the signal we want, ~ 0.9 μm at the BPM), and it modulates the optics everywhere — $\beta(s)$, $\phi(s)$, and Q . The pre-existing orbit (~ 0.37 mm rms, built by all 48 offsets together) is re-

transported through this slightly-changed optics, giving a few- μm response at the BPM: the optics “breathes.” Breathing beats signal by about 7 to 1. A clean decomposition makes the point: the response to modulating one quadrupole is -9.83 μm with the full pattern present, but only -0.037 μm when only that magnet’s own offset is kept — a factor 266. The measured amplitude reads *the orbit built by the neighbors*, not the magnet’s own offset, so dividing by a calibration is meaningless.

Because breathing is coherent (roughly the existing orbit times a number), it does not average away: going from 1 to 48 BPMs leaves the correlation at zero ($+0.07 \rightarrow -0.03$), and a SQUID’s low noise does not help, because noise was never the problem. Remove the breathing in an idealization and the same pipeline reconstructs perfectly (corr = 1.000), pinning the blame squarely on breathing. The C++ tracker confirms the analytic breathing amplitudes to $\sim 1\%$. BPM offsets are irrelevant in this branch — ac demodulation removes the static offset — so improving offsets from 100 to 30 μm changes nothing.

The two K -modulation failures are conceptually different: breathing is a coherent *contamination* that kills the per-magnet readout, while conditioning is a *noise blow-up* of the inversion. Different walls, and each closes its branch.

3. Neural-network reconstruction

Could a trained network beat the linear algebra? No, and the reason is instructive. The map misalignment $\rightarrow$ orbit is *exactly linear* ($y = R \Delta q$), so a network trained on it can only relearn R ; inverting is then R^{-1} with the same conditioning. In numbers, a multilayer perceptron (two hidden layers of 128, 4000 samples, 1 μm BPM noise) reconstructs the antisymmetric part as well as truncated SVD (0.71 vs. 0.63 μm error) and fails on the symmetric part just as badly (5.6 vs. 6.3 μm error against a 9.9 μm signal). The limit is information-theoretic: the symmetric imprint is below the noise-plus-systematics floor, and no estimator can pull out information that is not in the data. For the same reason fancier architectures cannot help — capacity is not the bottleneck in a linear problem — and for random noise the lock-in filter is already optimal, so learned time-series models cannot beat it either. Any real gain would need *prior information*; sparsity priors (LASSO, CLEAN, greedy harmonic selection) were tried and hit the same floor.

4. Calibrate once, then monitor drift (LOCO)

A pragmatic approach gives up on absolute alignment and asks only for its *changes*: calibrate the machine once (LOCO), then watch the orbit difference $y(t) - y_0$, in which the static offsets cancel. This works well for what it sees: drifts are tracked to 6.5 μm rms under 50 μm

Distributed-frequency K-modulation: optics breathing dominates the per-quad amplitude

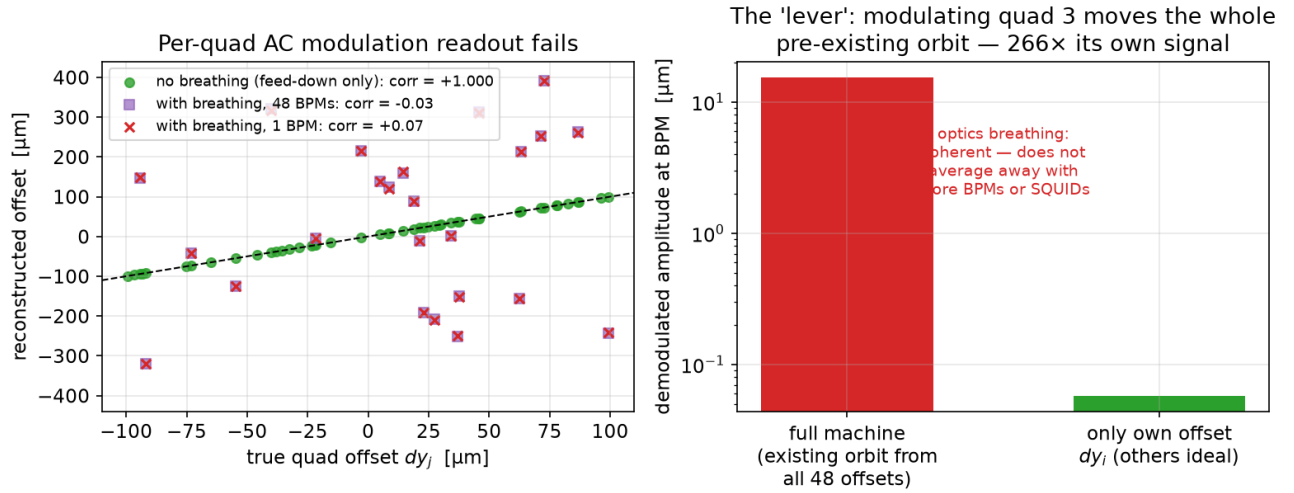


FIG. 5. Distributed-frequency per-quadrupole modulation. Left: reconstructed vs. true offsets — perfect without breathing (green), zero correlation with breathing, for 1 BPM (red) and even 48 BPMs (purple). Right: the “lever” — the demodulated amplitude responds to the whole pre-existing orbit ($266\times$ the magnet’s own contribution); being coherent, this contamination does not average away with more BPMs or a better (SQUID) resolution.

BPM offsets (a $29\times$ gain over absolute reconstruction), degrading only mildly with 1% β -beating. But the difference orbit still passes through the same response matrix: its eight worst modes are 96% symmetric with a $193\times$ noise blow-up. The drift monitor is a cheap, useful watchdog for the antisymmetric part — and blind to the symmetric part, like everything else in this class.

5. Harmonic approaches

Two harmonic variants deserve a note. *Targeted Fourier reconstruction*: if the harmonic content of the misalignment is known ahead of time, restricting the basis improves the conditioning from ~ 13000 to 186 and it works; but for unknown patterns, greedy or LASSO harmonic selection collapses from rank deficiency — knowing *which* harmonics is exactly the missing information. *Fit of a known signature*: reading a known field imprint (e.g., an $N=2$ stray radial field, which drives a $k=2$ vertical orbit) by projecting the 48-BPM data onto $\cos 2\theta/\sin 2\theta$ is perfectly conditioned but *biased*: the one $k=2$ number lumps together the field, the misalignment’s $k=2$ content, and the BPM offsets’ $k=2$ content — a ~ 53 nT bias at $10\ \mu\text{m}$ against a 1 nT target here. Full regularized SVD does not beat it under realistic offsets: its detection floor is ~ 45 nT at $100\ \mu\text{m}$ offsets (2.6 nT only for ideal, offset-free BPMs). In the language of Ref. [4], the forward projection is bias-limited, the inversion is noise-limited, and neither reaches the degenerate information.

6. The counter-rotating beam separation

Reference [1] proposes controlling the geometric phase by measuring and zeroing the *separation* between the CW and CCW orbits. But the separation is itself an orbit difference, so it inherits the orbit’s mode structure. We measured this directly [Fig. 6]: for $10\text{-}\mu\text{m}$ vertical patterns the ratio of antisymmetric to symmetric signature is $5.7\times$ in a single-beam orbit and $8.0\times$ in the CR separation (three seeds; an independent set gave $3.8\times$ and $4.5\times$). The suppression is similar, so *the CR separation opens no extra window on the symmetric part*. Measuring and zeroing it controls the visible part of the geometric phase — consistent with Ref. [1] — and leaves the symmetric residual of Table I untouched. This closes gap (ii) of the Introduction.

C. Trying to open the symmetric window at the lattice level

Two lattice-level escapes were examined.

1. Quadrupole polarity switching

Reversing all quadrupole currents ($g \rightarrow -g$) reverses each misalignment-driven spin rotation, so one might hope it flips the false EDM while leaving the real (electric-field-driven) EDM alone. It does not. The geometric phase is a *product of two* field-driven rotations, each $\propto g$, so f is *even* in g . Measured flip ratios $f(-g)/f(+g)$ are

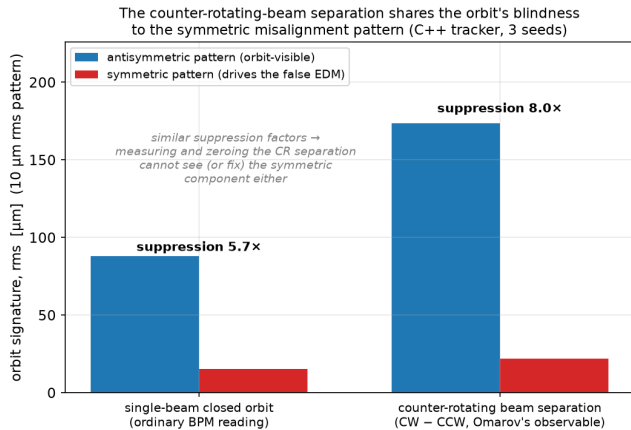


FIG. 6. The counter-rotating beam separation (right group) suppresses the symmetric pattern by about the same factor as the ordinary single-beam orbit (left group): the control observable of Ref. [1] shares the orbit’s blindness to the component that drives the residual false EDM.

+2.8, +0.85, and +1.17 for symmetric, antisymmetric, and generic patterns — the sign never reverses. Moreover, in the idealized periodic lattice the polarity flip is *degenerate* with reversing the beam: $f(\text{CCW}, +g) = f(\text{CW}, -g)$ to a measured 1.00, so the four-way combination of Eq. (C2) of Ref. [1] collapses to the two-way CW/CCW difference, and polarity switching gives no independent cancellation for the geometric phase. (It stays useful against systematics that are odd in g .) This closes gap (iii) of the Introduction.

2. Raising the betatron tune

The symmetric pattern is hidden because its kick harmonic ($k \approx 24$) sits far above the tune [Eq. (5)]; raising Q toward k should make it visible. It partly does: at $Q_y \approx 10.3$ (still alternating focusing, still stable) the high-harmonic part of the symmetric family reaches orbit gains like the antisymmetric ones. But the low-harmonic part needs $Q \geq 16$, while the 180° -per-cell stopband caps the tune at $Q \leq 12$; only ~ 8 –25% of the symmetric content becomes recoverable, worth only a factor ~ 2 in false-EDM suppression.

More decisively, a high tune needs *strong quadrupoles*: $Q_y : 2.3 \rightarrow 10.3$ needs $g : 0.21 \rightarrow 0.69$ T/m. Stronger gradients kick the beam harder for the same misalignment; since the geometric phase is a product of two such rotations ($\propto g^2$), plus a term from the larger betatron orbit, the false EDM grows steeply — a single-pattern C++ run gave a factor ~ 32 ($\sim g^3$) from $g = 0.21$ to 0.69 T/m — while the real EDM, driven by the electric field, is independent of g . So the signal-to-background *worsens* by that same large factor: one cannot operate at high tune, and a brief high-tune diagnostic transfers at most the recoverable $\sim 25\%$ back to the operating

point. (At $Q \approx 10$ the closed-orbit determination itself becomes delicate; our scans at $g = 0.69$ T/m carried residual-betatron contamination, so we quote only the robust qualitative conclusion and the single clean pattern above.)

D. The one technique that escapes the wall: null-finding BBA

All the walls above come from either inverting a response matrix or dividing a demodulated amplitude by a calibration. Standard quadrupole beam-based alignment (BBA) does neither. It steers the beam through each quadrupole and finds the magnetic center as the beam position where the response to a gradient change *vanishes* — a zero-crossing. A zero-crossing does not depend on how big the response is, so it is not amplified by the conditioning, and it is transparent to β -beating in ideal optics. It carries no “symmetric blindness.”

a. Idealized optics: it works. In clean optics, null-finding BBA measures the symmetric centers and cuts the false EDM from $\sim 356\times$ to $\sim 1.6\times$ target — the wall of Sec. III B is genuinely absent for this method.

b. Realistic optics: breathing hits here too. Under 1% β -beating a single pass fails, for a familiar reason. To steer the beam through one magnet we build a local orbit bump, and modulating that magnet re-transportes the large uncorrected orbit through the beating optics (the same breathing as Sec. III B 2), biasing the zero-crossing. The cure is the natural one: correct the orbit first so it is small, then repeat — “align, flatten, re-align.” Each pass shrinks the orbit, so each pass breathes less.

c. What the completed run shows. We ran the full iteration in the C++ tracker with 1% β -beating, three passes (Table II). The vertical plane behaves exactly as hoped: the orbit shrinks $48 \rightarrow 6.9 \mu\text{m}$, the vertical symmetric residual falls roughly by half each pass ($3.3 \rightarrow 1.7 \rightarrow 0.9 \mu\text{m}$), and it is heading for the clean-optics floor. The horizontal plane does not: its symmetric residual sticks near $2.4 \mu\text{m}$. Because the false EDM is the product $dx \cdot dy$, the falling vertical part still pulls it down — from $356\times$ to $145\times$ target over the three passes — but the stuck horizontal part sets a floor well above the goal.

d. Why vertical works and horizontal stalls. This is the honest crux, and it is not a law of physics — it is a limitation of the *model we steer with*. The local bumps are built from the analytic optics solver of Sec. II, which treats the electric deflectors as field-free drifts. In the vertical plane that is nearly exact: the ring’s radial deflector field barely focuses vertically, so the model matches the real machine, the bumps land where intended, and the vertical BBA reaches its floor. The horizontal plane is the plane the ring bends in, and there the deflectors do two things the model omits — they focus the beam horizontally, and they bend it, which creates dispersion (a coupling between momentum and position). With

TABLE II. Iterated null-finding BBA under 1% β -beating (C++ tracking, three passes). “sym” is the symmetric-component residual of Eq. (4). The vertical residual falls toward the clean floor; the horizontal one sticks, and the product $dx \cdot dy$ sets the false-EDM floor.

pass	orbit [μm]	sym dy [μm]	sym dx [μm]	$ f $ / target
raw	48	—	—	356×
1	24.1	3.30	4.48	368×
2	12.6	1.72	2.99	238×
3	6.9	0.91	2.38	145×

both missing from the steering model, the bumps are mis-sized, worst at the few locations where the accumulated model error is largest (near the start of the lattice), and there the zero-crossing is biased and the correction stalls. A one-line check confirms the cause. Because the model treats the deflectors as drifts, it predicts *identical* tunes in the two planes, $Q_x = Q_y = 2.303$. The real ring matches this in the vertical plane to four decimals (0.3032 measured vs. 0.3032), but its horizontal tune is higher by 0.052 in fractional part — the horizontal focusing of the deflectors, absent from the model. Since the deflectors are the *only* element that distinguishes the two planes, this pins the horizontal stall on the model, not on physics; even in perfectly clean optics (no β -beating) the same few horizontal magnets are already mis-measured while the vertical ones are not. The standard fix is exactly what real machines do — build the bumps from the *measured* horizontal response of the ring rather than an idealized model — and is left for future work.

e. Two knobs, one trade-off. The iteration has two settings: how aggressively to apply each correction, and a quality cut that throttles magnets whose zero-crossing is inconsistent across BPMs. Applied at full strength with no quality cut, the horizontal plane diverges (its residual blows up to $\sim 12 \mu\text{m}$). With under-relaxation and the quality cut, the divergence is stopped ($2.4 \mu\text{m}$), but the throttled horizontal magnet is simply never corrected, so the false EDM plateaus at 145× rather than reaching the target. The quality cut buys *stability*, not a *solution*: the real fix is the better horizontal model above, not a smarter iteration.

f. Summary of the method. Null-finding BBA is the one measurement class not excluded by the walls of Sec. IIIB. In clean optics it recovers the symmetric component outright; under realistic optics it already solves the vertical plane, and the horizontal plane is a well-understood, fixable modeling gap rather than a fundamental limit. We present it as the promising open route, not as a finished solution — the horizontal model and the hardware floor below remain to be done.

E. Outside the orbit: off-momentum spin sensing (brief)

For completeness we record one attempted escape outside the orbit toolbox. Storing the beam slightly off the magic momentum makes the in-plane spin precess at $\nu_s \neq 0$, and the geometric-phase part of the vertical oscillation is indeed *amplified* $\propto 1/\nu_s$ — suggesting a fast, resonant readout. It fails because misalignments also tilt the invariant spin axis at *first* order in σ : this $\mathcal{O}(\sigma)$ oscillation sits at the same frequency ν_s and beats the $\mathcal{O}(\sigma^2)$ geometric term by ~ 3600 at $\sigma = 10 \mu\text{m}$. Every way to separate the two was tested and closed: they do not separate in $\delta p/p$, their relative phase is pattern-dependent, four-fold sampling does not cancel the first-order term (it is common to the four particles), the amplitude crossing lies at an unphysical $\sigma \approx 62 \text{ mm}$, a full 3D fit of the invariant axis leaves no residual, and the longitudinal component is knob-insensitive off-magic. The magic momentum is the *unique* working point that turns the dominant first-order effect into a harmless constant offset — which is also why the estimator of Sec. IIB works there.

IV. CONCLUSIONS

a. Quantitative result. The orbit-based chain — both beams at once, standard orbit correction, and continuous drift monitoring — bounds the quadrupole-misalignment false EDM at $\sim 6 \times 10^{-28} \text{ ecm}$ for $\sigma = 10 \mu\text{m}$, and the bound scales as σ^2 (Fig. 3). If the required suppression were only to 10^{-27} ecm , existing BBA technology would suffice. Reaching 10^{-29} ecm needs the *symmetric* part of the magnetic-center misalignment held near $1.3 \mu\text{m}$.

b. Wall for the response-matrix and amplitude classes. The symmetric component is unobservable at the required level in every channel of these classes we tested: common-frequency K -modulation with (regularized) inversion, per-quadrupole modulation read as an amplitude, neural networks, LOCO-plus-drift monitoring, harmonic fits, and the counter-rotating beam separation. For all of them the limit is set not by random BPM noise — which synchronous detection beats — but by two coherent effects that do not average down: optics breathing under per-magnet modulation, and β -beating amplified by the conditioning of the inversion; and by the lattice itself, whose stopband caps the tune far below the symmetric kick harmonic. *For these classes no BPM technology, including SQUID BPMs, moves the floor.* Polarity switching gives no independent cancellation (it is degenerate with beam reversal, and the geometric phase is even in g), and a tape measure cannot substitute, because the driving quantity is the beam-defined magnetic center.

c. The one class that escapes: null-finding BBA. Standard BBA locates each magnetic center by a zero-

crossing rather than by inverting a matrix or dividing an amplitude, so the walls above do not apply. In clean optics it recovers the symmetric component and cuts the false EDM from $356\times$ to $1.6\times$ target. Under realistic 1% β -beating, a single pass is spoiled by the same optics breathing, but flattening the orbit first and iterating recovers the *vertical* plane cleanly ($0.9\ \mu\text{m}$ residual and falling; false EDM $356\times \rightarrow 145\times$ over three passes). The *horizontal* plane stalls, but for an understood and fixable reason: our steering model treats the electric deflectors as drifts, which is accurate vertically but not in the bending (horizontal) plane, where the deflectors focus and create dispersion. Building the horizontal bumps from the machine’s measured response — ordinary BBA practice — is the natural next step. We therefore present null-finding BBA as the one route *not* excluded by the walls above: promising, partly demonstrated, and not yet finished (the horizontal model and the hardware floor below remain).

d. The hardware floor. One caveat applies to any modulation-based method, including null-finding BBA: changing a magnet’s gradient may also move its magnetic center a little, through hysteresis or heating. On a real machine this motion can be at the micron level and would sit directly on top of the signal. We have not simulated it; it is a hardware-metrology question and, together with the horizontal model, is the main thing standing between the present result and a complete method.

e. Scope. Active spin-based nulling is out of scope for a basic reason: the false and real EDM share the observable dS_y/dt , so a feedback that nulls the measured precession nulls the signal too; the only discriminator is the CW/CCW difference, which is exactly what leaves the $62\times$ residual. (Its statistics are also prohibitive: po-

larimetry reaches nrad/s only on year timescales [10].) Forward-predicting the false EDM from the measured orbit — as opposed to reconstructing misalignments — is a separate open problem not covered here.

f. Implications for the experiment. Our results support the control philosophy of Ref. [1] while sharpening its edges. (a) The CW/CCW combination and orbit-based separation control handle the visible part of the geometric phase, and a standard-BPM orbit monitor is a cheap substitute for SQUID pickups for that part (a $7.7\times$ gain), with drift monitoring as a free continuous watchdog. (b) The symmetric residual must be controlled *by construction* — through magnetic-center alignment tolerances and field quality — because it can be neither measured by the matrix/amplitude classes nor actively nulled afterwards; null-finding BBA is the one measurement that may yet reach it, pending the horizontal model and the hardware floor. (c) The σ^2 law makes the tolerance concrete: every factor of two in symmetric alignment buys a factor of four in false EDM.

g. Validity. All numbers are for one lattice (24 FODO cells, $Q_y \approx 2.3$) in a linear model (no sextupoles); the suppression factors ($3.4\times$, $7.7\times$, $62\times$) carry finite ensemble statistics; and the quadrupole roll angle is a separate direct channel with a ~ 0.3 mrad tolerance, outside the misalignment budget here. The qualitative structure — the symmetric/antisymmetric visibility gap and the coherent-systematic walls — follows from the alternating-gradient principle itself and should carry over to any strong-focusing ring.

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